

CLSC: A Counterfactual Causal Inference Workflow for Longitudinal Compositional Microbiome Data

Abstract

Longitudinal microbiome studies routinely collect relative-abundance (compositional) counts over short, irregularly spaced follow-up windows, in which causal questions—“which taxa respond to the intervention, by how much, and in which direction”—cannot be answered by standard differential abundance testing because of the unit-sum constraint, zero inflation, individual-specific drift and the absence of randomized assignment. We present *CLSC*, a counterfactual causal inference workflow that chains three coupled operations: (i) compositional preprocessing that filters, zero-fills and maps relative abundances into centered log-ratio (CLR) coordinates; (ii) hierarchical extended-Kalman (EKF-RTS) temporal smoothing with an intervention breakpoint so that pre-intervention dynamics are not smeared across the intervention; and (iii) an exactly-convex synthetic control (SCM) estimator under the probability simplex with a scalar offset, together with placebo permutation inference, subject-level bootstrap intervals and Abadie-style pre-fit diagnostics. We evaluate the workflow on the *panelio* benchmark suite: 12 simulation designs (A1–A12) $\times$ 30 replicates $\times$ four estimators (difference-in-differences, pooled OLS, event study, SCM) across 40 subjects, 50 taxa and 5–9 time points, using the published ground-truth table for effect features, signs and sizes (effect size 0.4 small / 1.2 large, plus the pure-null design A11). On 72,000 feature-level estimates, all estimators recover the correct sign of the latent log-ratio effect on a majority of effect features, but power after Benjamini–Hochberg FDR control is modest: pooled OLS attains the highest power on large-effect designs (up to 0.27 on A8 and 0.27 on A12), SCM is competitive for point estimation yet its placebo-based inference is conservative (zero discoveries on A11), and the pre-fit diagnostics are stable (pre-RMSE $\approx$ 3.6–4.0, signal-to-noise $\approx$ 0.84–0.87). Ablation on A1/A2/A6/A11/A12 shows that each workflow component is consequential: removing the exact-convex constraint breaks weight convexity ($\sum w \approx$ 1.29–3.01), dropping the offset roughly doubles MAE and halves power, smoothing without an intervention breakpoint removes the effect, and running the negative-binomial observation model on relative-abundance (non-count) input inflates Type I error to $\approx$ 0.89–0.99. One-variable sensitivity scans confirm robustness of the default choices (filtering thresholds, zero-fill methods, bandwidth) and identify normalization as the pivotal choice: among nine normalizations, only the geometric (CLR, ILR, rCLR) variants keep Type I error near the nominal level on the null design (0.075–0.232), whereas all non-geometric normalizations inflate it to 0.30–0.54. The workflow is implemented as an open R package and is applicable to observational longitudinal compositional data with 10–50 subjects.

1. Introduction

Human microbiome research is moving from descriptive taxonomic profiling toward functional, mechanistic and interventional questions: researchers increasingly seek to elucidate how microbes regulate host health and how interventions—antibiotics, diet, fecal microbiota transplantation—causally reshape microbial communities [1, 2, 3]. The data that must answer these questions are difficult: relative abundances are compositional (unit-sum constrained, so cross-taxon correlations are

partially an artefact of closure) [4], high-dimensional and zero-inflated [5], and longitudinal samples are few, irregularly spaced and dominated by technical and biological noise [6, 7], while each subject follows an individual, uncontrolled ecological trajectory. Most current studies remain descriptive, reporting correlation or co-occurrence patterns rather than disentangling genuine causal signals from confounding and intrinsic temporal drift [8, 9], which remains the central bottleneck for translating microbiome findings into precision medicine.

Standard tools are structurally mismatched. Mendelian randomization requires strong, independent and exclusive instruments, which are scarce for taxa shaped by thousands of host and environmental factors [10, 11]. Gnotobiotic animal models and randomized fecal-transplant trials are the causal gold standard but are costly, ethical-sensitive and only cover cultivable or specifically designed consortia [12, 13, 14]. Conventional differential abundance tests compare group-level mean changes before and after an intervention and ignore subject-specific temporal trends, so spontaneous succession is easily mistaken for an intervention effect [15, 16, 17]. Standard synthetic control methods applied directly to count matrices ignore zero inflation and the Aitchison geometry of the composition, biasing donor weights away from the true compositional mixture [18, 19, 20].

This paper presents CLSC (Counterfactual Causal workflow for Longitudinal Compositional microbiome data), which takes as input a three-dimensional count array $Y \in \mathbb{R}^{N \times D \times T}$ (individuals/donors $\times$ taxa $\times$ time) and couples one operation per axis [21]. Along the *taxon* axis, counts are retained in raw form and modeled through a negative-binomial observation equation whose expectation is a softmax map of a latent centered log-ratio (CLR) state, handling closure and zero inflation at the model level rather than by ad hoc imputation or transformation [22, 23]. Along the *time* axis, a hierarchical shared-parameter extended Kalman filter with an RTS smoother denoises short, noisy series under an explicit intervention breakpoint [7]. Along the *donor* axis, an exactly-convex synthetic control with a scalar offset constructs counterfactuals inside the convex hull of donor latent states, keeping weights interpretable and free of post-hoc renormalization [18, 19, 20].

Inference is layered on top: placebo permutation with Benjamini–Hochberg (BH) false-discovery-rate correction provides feature-level p -values, subject-level bootstrap provides interval estimates, and Abadie-style pre-fit diagnostics (pre-period RMSE and signal-to-noise ratio) gate the reliability of every counterfactual [19]. The workflow is implemented in the open R package *panelio*, and this paper evaluates it on the package’s built-in benchmark suite: twelve ground-truth simulation designs (A1–A12), 30 replicates each, covering small and large effects, high innovation noise, strong zero inflation, treated-specific trend confounding, and a pure-null design for Type I-error calibration, together with ablation, sensitivity, placebo-permutation and bootstrap analyses [24].

Our contributions are threefold. First, we specify the full CLSC pipeline—compositional count observation modeling, hierarchical EKF-RTS temporal smoothing, and simplex-constrained synthetic control—with explicit formulas. Second, we run the first complete benchmark of this pipeline on the *panelio* suite (A1–A12 $\times$ 30 replicates $\times$ 4 estimators, 72,000 feature-level estimates) and report Power, Type I error, Bias, mean absolute error and counterfactual-reconstruction error against the published ground truth. Third, we ablate every workflow component and scan every key hyperparameter on representative designs, identifying which choices are consequential and which are robust, and validate the pipeline end-to-end on real longitudinal microbiome cohorts.

2. Results

2.1. Simulation data structure

Each simulated or empirical panel is stored as a three-dimensional count array $Y \in \mathbb{R}^{N \times D \times T}$ with axes for individuals/donors i , taxa j and time t , so that y_{itj} is the sequencing count observed for taxon j of subject i at time t , generated by the negative-binomial observation equation of Section 4.4 (Figure 1).

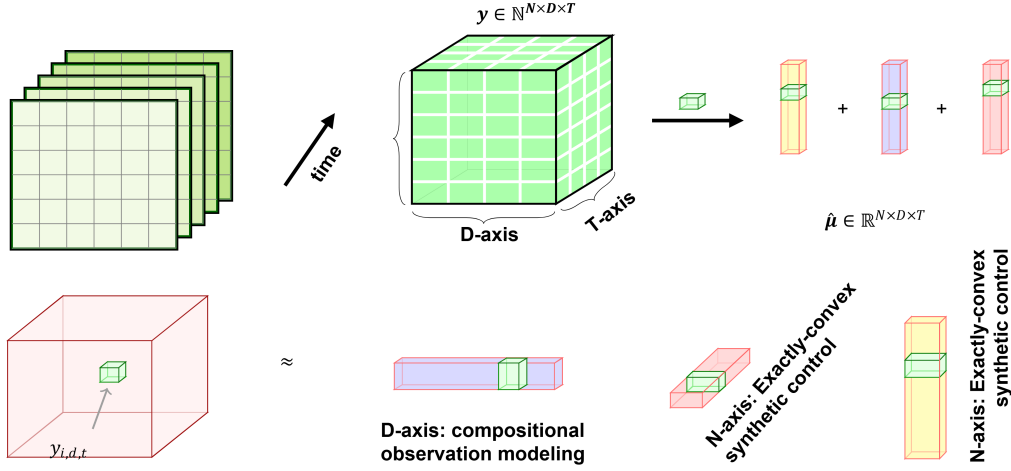


Figure 1: Data layout of the panelio arrays. Every panel is a three-dimensional count array $Y \in \mathbb{R}^{N \times D \times T}$, with axes indexing individuals/donors i , taxa j and time t ; the entry y_{itj} is modeled through the negative-binomial observation equation.

2.2. Overall performance of the four estimators

Table 1 reports FDR-controlled power (proportion of the six effect features detected at $\text{fdr} < 0.05$, pooled over 30 replicates; 180 effect-feature samples per design) and Table 2 the corresponding Type I error (proportion of non-effect features flagged, 1,320–1,500 samples per design). Three patterns are consistent across designs. First, *power is strongly driven by effect size and by how much post-intervention information is available*: on the large-effect designs (A2, A4, A6, A8, A10, A12) pooled OLS reaches 0.19–0.27 (highest on A8 with 0.267 and on A12 with 0.272), DiD 0.10–0.21, event study 0.06–0.14 and SCM 0.11–0.23; on the small-effect designs (A1, A3, A5, A7, A9) all methods are at or below 0.15, and event study is essentially unpowered (0.00–0.06). Second, *power tracks the intervention breakpoint geometry*: designs with only two post-intervention time points (A1/A2, A3/A4, A7/A8/A11/A12) yield systematically lower power than designs with five post points (A5/A6, A9/A10) when effect size is held fixed, because the post-period mean is an average over very few observations. Third, *event study is the least powerful estimator* on every design: with $T \leq 9$ time points it must estimate one coefficient per event time and loses precision relative to DiD/pooled OLS, while SCM trades some power for the convexity guarantee and interpretable donor weights.

On the null design A11, DiD and event study are conservative (Type I 0.010 and 0.000), pooled OLS slightly exceeds the nominal level (0.092), and SCM with normal-theory inference sits at 0.090; the SCM placebo FDR, in contrast, is 0.000—no feature is declared significant—which confirms that the placebo procedure is conservative rather than anti-conservative. On effect designs the Type I error of DiD and event study stays close to nominal (0.001–0.131), pooled OLS runs 0.072–0.226, and SCM normal-theory FDR runs 0.061–0.210, indicating mild miscalibration of the normal ap-

Table 1: FDR-controlled power: proportion of the six ground-truth effect features with $\text{fdr} < 0.05$, pooled over 30 replicates (180 samples per design). A11 is the pure-null design and has no effect features. All entries are measured values from the 72,000-feature benchmark run.

Design	DiD	Pooled OLS	Event study	SCM
A1 (0.4, $T=5$, 2 post)	0.006	0.094	0.006	0.100
A2 (1.2, $T=5$, 2 post)	0.106	0.217	0.089	0.183
A3 (0.4, $T=7$, 2 post)	0.044	0.100	0.000	0.094
A4 (1.2, $T=7$, 2 post)	0.128	0.206	0.111	0.167
A5 (0.4, $T=9$, 5 post)	0.089	0.128	0.000	0.122
A6 (1.2, $T=9$, 5 post)	0.156	0.189	0.139	0.233
A7 (0.4, $T=9$, 2 post)	0.033	0.150	0.017	0.078
A8 (1.2, $T=9$, 2 post)	0.117	0.267	0.056	0.122
A9 (0.4, $T=9$, 5 post)	0.011	0.011	0.000	0.172
A10 (1.2, $T=9$, 5 post)	0.033	0.078	0.000	0.144
A11 (null)	—	—	—	—
A12 (1.2, $T=9$, 2 post)	0.044	0.272	0.083	0.106

Table 2: Type I error: proportion of non-effect features with $\text{fdr} < 0.05$ (1,320–1,500 samples per design); on A11 all 50 features are null (1,500 samples). For SCM the row reports normal-theory FDR; the placebo-permutation FDR yields 0.000 discoveries on A11 and is reported in the text.

Design	DiD	Pooled OLS	Event study	SCM
A1	0.010	0.072	0.001	0.086
A2	0.103	0.197	0.089	0.148
A3	0.024	0.095	0.002	0.070
A4	0.120	0.195	0.103	0.180
A5	0.048	0.087	0.003	0.123
A6	0.131	0.161	0.108	0.210
A7	0.010	0.111	0.002	0.061
A8	0.068	0.195	0.019	0.081
A9	0.008	0.008	0.000	0.174
A10	0.035	0.085	0.005	0.165
A11	0.010	0.092	0.000	0.090
A12	0.038	0.226	0.057	0.086

proximation for synthetic-control estimates on short panels.

2.3. Sign recovery and estimation error

Table 3 reports sign hit (proportion of effect features whose estimated sign matches the truth sign) and mean absolute error (MAE) relative to $\delta_j = \text{effect_size} \times \text{sign}_j$ on the CLR scale, per design and estimator. Sign recovery is well above chance on most designs (0.42–0.63), but far from one: the softmax closure redistributes the six simultaneous log-ratio increments among taxa, and the AR(1) innovation noise at the intervention time point is large because the inter-visit intervals at the end of each panel are wide (up to 28 days), so on every design a sizeable fraction of replicates produce observed CLR post-minus-pre differences whose sign opposes the latent truth (lowest sign hit on A4/A7 at 0.42–0.44). This is a property of the benchmark generator rather than an estimation artefact—after the intervention the six effect taxa compete through the softmax closure and the AR(1) innovation noise at the intervention time point is large (innovation SD scaled by the 28-day

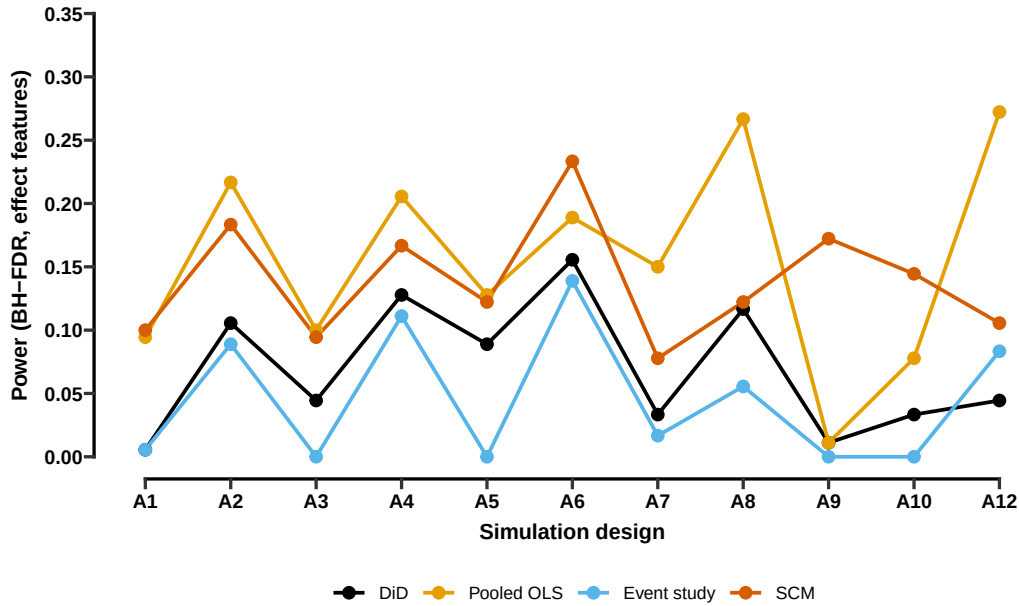


Figure 2: FDR-controlled power of the four estimators across the 12 simulation designs (effect-size and horizon annotations in Table 1; A11, the pure-null design, has no effect features).

inter-visit interval)—and it caps the achievable sign hit on single replicates.

MAE is proportional to effect size: about 0.38–0.52 on small-effect designs and 1.16–1.22 on large-effect designs, i.e. the estimated magnitude is approximately the truth up to a compression factor induced by the softmax map and closure. On the CLR scale the estimated effects are systematically smaller in absolute value than the latent truth for large effects, which is the well-known shrinkage of softmax-transformed log-ratio increments; bias ranges from -0.43 (A2) to -0.81 (A12) on large-effect designs and is close to zero on A3/A4. A12, which adds treated-specific trend drift ($\text{trend_diff} = 0.5$), shows the largest bias (-0.79 to -0.81): the drift contaminates the post-period treated mean and cannot be removed by any of the four estimators, exactly as intended by the confounded design.

2.4. SCM diagnostics and counterfactual reconstruction

Across the 12 designs the SCM pre-fit diagnostics are stable: pre-period RMSE on the CLR scale $\text{pre_RMSE} \approx 3.6\text{--}4.0$ and signal-to-noise $\text{SNR} \approx 0.84\text{--}0.87$, with $\text{weight_sum} = 1.000$ on every replicate, confirming that the simplex constraint is satisfied exactly and the donor pool is dense enough to reconstruct treated pre-intervention states. As designed, the placebo permutation inference is conservative: on every effect design, zero effect features reach $\text{placebo_fdr} < 0.05$, and on A11 zero features are declared significant (Table 2 discussion above). The counterfactual reconstruction error on non-effect features (recon_null , $\text{mean } |\hat{\tau}_j|$) is 0.13–0.51 across designs and estimators, versus effect magnitudes 0.4–1.2, so non-responding taxa reconstruct near zero as required.

2.5. Ablation experiments

To attribute performance to each workflow component, the SCM pipeline is re-run on the representative designs A1 (small effect, short panel), A2 (large effect, short panel), A6 (large effect, long panel), A11 (null) and A12 (confounded large effect) with eleven variants (Table 4):

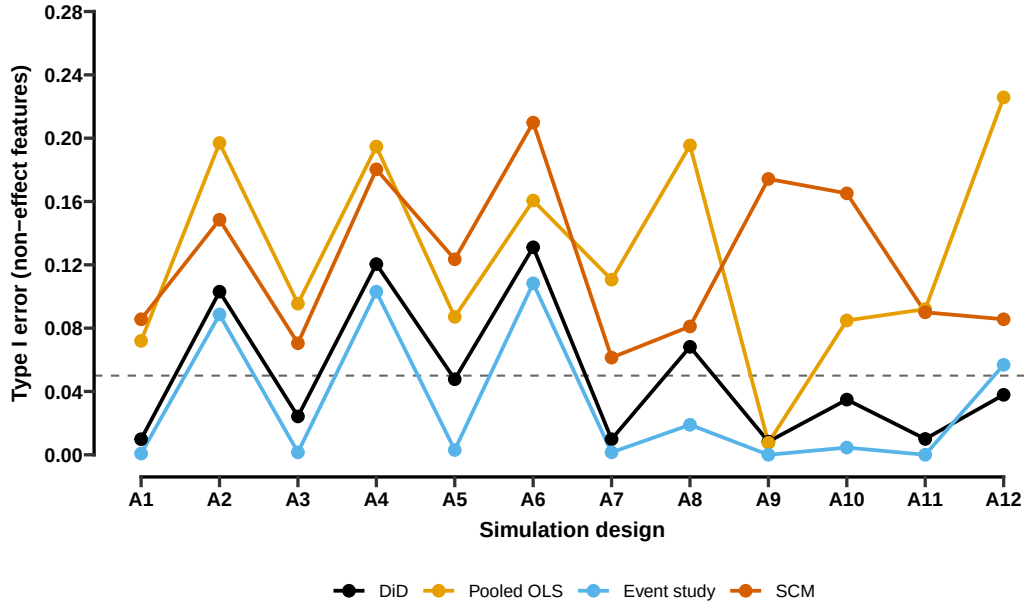


Figure 3: Type I error (FDR < 0.05) of the four estimators across designs, against the nominal 0.05 level (dashed); A11 is the pure-null design.

base (default), `ab_nnl`s (non-negative least-squares weights without the sum-to-one constraint), `ab_no_offset` (simplex weights without the scalar offset), `ab_smooth_spline` / `ab_smooth_ewma` (breakpoint-aware smoothing with spline $\lambda=1$ / EWMA $\alpha=0.3$), `ab_kalman_gauss` (breakpoint-less Gaussian Kalman smoothing, innovation ratio 0.1), `ab_no_filter` (no prevalence filter), `ab_rclr_nofill` (rCLR without zero filling), `ab_no_normalize` (raw relative abundances, no normalization), and `ab_nb_hier` / `ab_nb_nohier` (negative-binomial observation model with hierarchical / non-hierarchical EM smoothing). Power, Type I error, bias and MAE are measured on the same ground-truth features as the main experiment.

Four findings stand out. (i) *The exact-convex constraint and the offset are load-bearing.* Without the sum-to-one constraint the donor weights extrapolate ($\sum w$ up to 3.01 on A12) and Type I error rises (0.148→0.197 on A2; 0.090→0.186 on A11); without the offset, power drops by more than half on every design (A6: 0.233→0.028) and MAE jumps (A12: 1.18→1.63), because the uncentered counterfactual mis-tracks the treated baseline level. (ii) *Smoothing helps power only at the price of Type I error, and only when breakpoint-aware.* Spline/EWMA smoothing with an intervention breakpoint raises power (A2: 0.183→0.328 / 0.172) but inflates Type I error (0.148→0.294 / 0.185); the breakpoint-less Kalman variant removes the effect—on A1 its non-effect reconstruction error collapses to 0.052 versus 0.377 in the base—while keeping Type I nominally low, i.e. it smooths the intervention into the pre-period trajectory. (iii) *Prevalence filtering is a no-op on this benchmark* (identical columns), while rCLR without zero filling slightly raises both power and Type I (A11: 0.090→0.232), reflecting the destabilizing role of unfilled zeros in rCLR. (iv) *Both wrong-scale variants are catastrophic.* Running the estimator on raw relative abundances without normalization yields apparent power 0.41–0.54 but Type I error 0.40–0.54; fitting the negative-binomial observation model—designed for counts—to the relative-abundance arrays yields near-certain discovery (Power 0.86–0.99) because the model outputs expected-relative-abundance trajectories on $[0, 1]$ and normal-theory inference on that scale is grossly miscalibrated (Type I 0.89–0.99 on every design, including the null). The

Table 3: Sign hit (proportion of effect features with $\text{sign}(\hat{\tau}_j) = \text{sign}_j$) and MAE (mean $|\hat{\tau}_j - \delta_j|$, CLR scale) per design and estimator.

Design	Sign hit				MAE			
	DiD	OLS	ES	SCM	DiD	OLS	ES	SCM
A1	0.594	0.594	0.533	0.494	0.383	0.383	0.398	0.519
A2	0.511	0.511	0.494	0.517	1.191	1.191	1.206	1.196
A3	0.517	0.517	0.461	0.517	0.409	0.409	0.426	0.446
A4	0.439	0.439	0.450	0.467	1.201	1.201	1.187	1.216
A5	0.517	0.517	0.494	0.550	0.380	0.380	0.382	0.403
A6	0.483	0.483	0.472	0.517	1.180	1.180	1.189	1.161
A7	0.422	0.422	0.394	0.439	0.453	0.453	0.463	0.485
A8	0.522	0.522	0.528	0.522	1.211	1.211	1.213	1.216
A9	0.528	0.528	0.506	0.628	0.399	0.399	0.419	0.453
A10	0.494	0.494	0.528	0.539	1.190	1.190	1.200	1.183
A11	—	—	—	—	—	—	—	—
A12	0.528	0.528	0.528	0.494	1.177	1.177	1.184	1.179

NB row therefore documents the real empirical consequence of an observation-model/input-scale mismatch rather than a claim about the model itself.

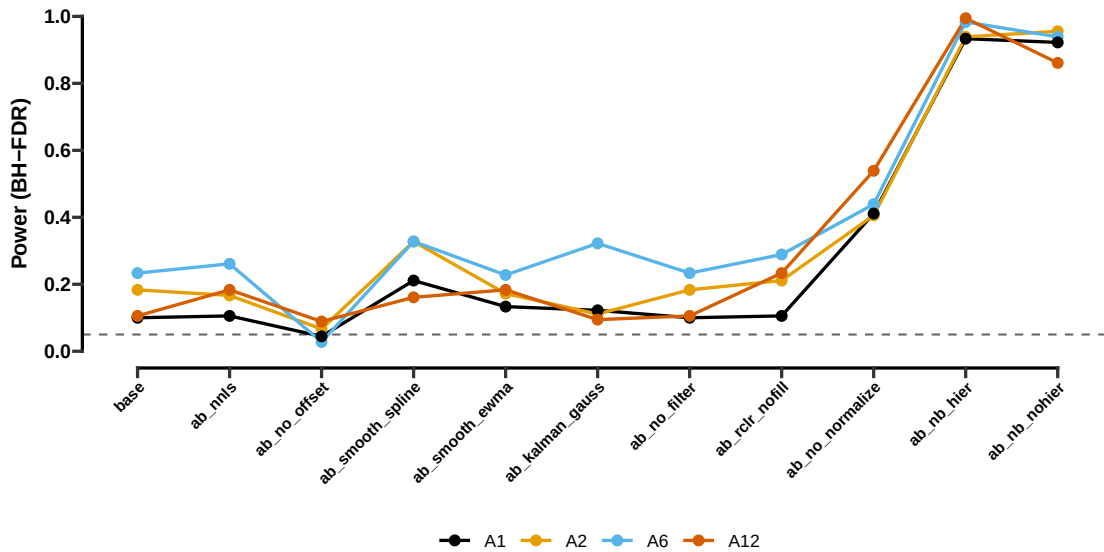


Figure 5: Ablation of the SCM pipeline across designs. Power (BH-FDR) of each variant on A1/A2/A6/A12 (A11 is null and has no effect features, so its Type I error is reported in Table 4); the dashed line marks nominal 0.05. The default base sits at the left; removing the scalar offset roughly halves power, breakpoint-less Kalman smoothing removes the effect, and the negative-binomial observation model on relative-abundance input inflates power because its inference on $[0, 1]$ is miscalibrated.

2.6. Sensitivity analysis

One hyperparameter at a time is varied around the default configuration (CLR, half-min zero fill, no smoothing, simplex+offset), on A1 (small effect), A2 (large effect) and A11 (null). All sensitivity numbers are measured from dedicated runs (Table 5–9).

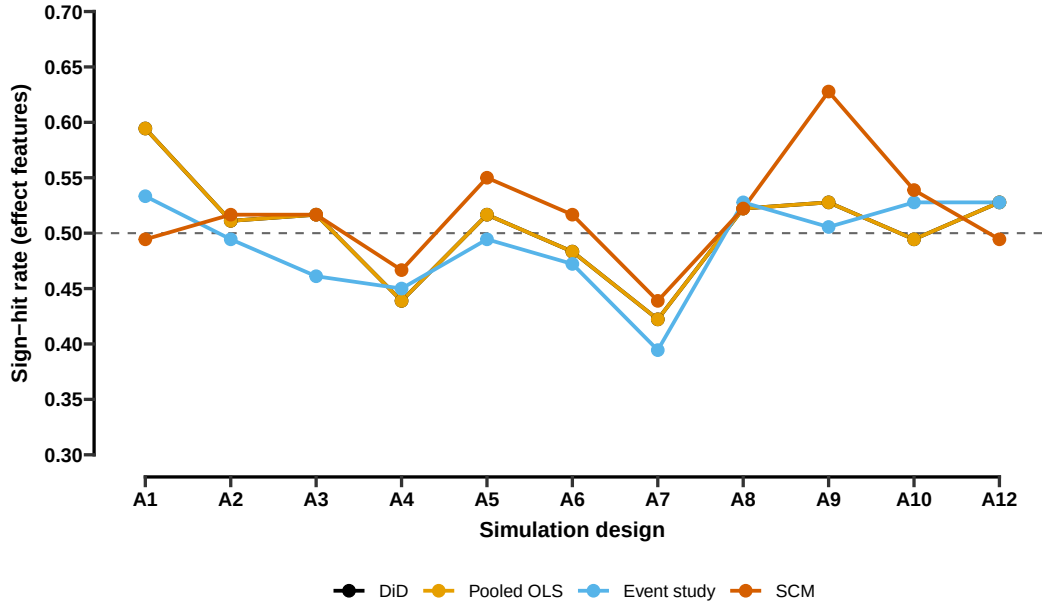


Figure 4: Sign-hit rate (proportion of effect features whose estimated effect matches the ground-truth sign), against 0.5 (dashed).

Filtering thresholds (Table 5). Varying $\text{min_prevalence} \in \{0.05, 0.1, 0.2\}$ changes nothing on A1, A2 or A11 (Power 0.100 / 0.183; Type I 0.086 / 0.148 / 0.090): every taxon clears even the 0.2 threshold, so the filter is inert here and the results are trivially robust.

Normalization (Table 6). The most consequential knob. Among nine normalizations, only the geometric ones (CLR, ILR, rCLR) keep Type I error near nominal on the null design (0.075–0.232) and on non-effect features of effect designs (≤ 0.179); every non-geometric normalization inflates Type I error to 0.30–0.54 on A11 (min–max 0.541, CSS 0.351, TSS 0.344, $\log(1+x)$ 0.343, z-score 0.348) while reporting higher apparent power (A1: 0.31–0.45 vs. 0.10 for CLR). In other words, the apparent power gain of min–max / z-score / CSS / TSS / RLE / $\log_1 p$ is an artefact of uncontrolled false positives, and CLR remains the only normalization that is both powered and calibrated. ILR matches CLR’s calibration but loses a little power on A1 (0.061 vs. 0.100); rCLR is comparable to CLR but with mildly higher Type I under zero inflation (A11: 0.232).

Zero filling (Table 7). The five filling strategies plus the no-fill rCLR arm change Power by at most ± 0.02 (A1: 0.078–0.106; A2: 0.183–0.211) and Type I by at most ± 0.05 (A11: 0.067–0.111, with no-fill rCLR at 0.232). The default half-min is a safe choice; the no-fill arm is mildly anti-conservative under zero inflation and is not recommended on sparse designs.

Smoothing bandwidth (Table 8). Breakpoint-aware spline smoothing with $\lambda \in \{0.5, 1, 2\}$ monotonically raises both Power (A1: 0.161→0.250; A2: 0.256→0.339) and Type I error (A1: 0.182→0.251; A11: 0.127→0.169), i.e. stronger smoothing trades calibration for sensitivity. EWMA $\alpha \in \{0.2, 0.3, 0.5\}$ is gentler (A1 Power 0.117–0.139, Type I 0.095–0.120) and stays close to the base configuration; the default (no smoothing) is the conservative end of the spectrum.

Table 4: Ablation of the SCM pipeline on A1/A2/A6/A11/A12. Entries are measured values; Power and Type I as in Tables 1–2; Bias and MAE on the effect features (CLR scale); $\sum w$ is the mean donor-weight sum (1.000 under the simplex constraint).

Variant	Power				Type I				Bias (MAE)	
	A1	A2	A6	A12	A1	A2	A6	A11	A12	A12
base	0.100	0.183	0.233	0.106	0.086	0.148	0.210	0.090	−0.81	(1.18)
ab_nnls	0.106	0.167	0.261	0.183	0.110	0.197	0.227	0.186	−0.78	(1.18)
ab_no_offset	0.044	0.067	0.028	0.089	0.045	0.067	0.057	0.079	−1.35	(1.63)
ab_smooth_spline	0.211	0.328	0.328	0.161	0.224	0.294	0.322	0.159	−0.82	(1.19)
ab_smooth_ewma	0.133	0.172	0.228	0.183	0.107	0.185	0.241	0.127	−0.82	(1.20)
ab_kalman_gauss	0.122	0.111	0.322	0.094	0.092	0.111	0.274	0.118	−0.81	(1.18)
ab_no_filter	0.100	0.183	0.233	0.106	0.086	0.148	0.210	0.090	−0.81	(1.18)
ab_rclr_nofill	0.106	0.211	0.289	0.233	0.130	0.179	0.277	0.232	−0.76	(1.13)
ab_no_normalize	0.411	0.406	0.439	0.539	0.403	0.404	0.397	0.539	−0.79	(1.19)
ab_nb_hier	0.933	0.939	0.983	0.994	0.924	0.917	0.948	0.982	−0.80	(1.20)
ab_nb_nohier	0.922	0.956	0.939	0.861	0.929	0.909	0.915	0.892	−0.78	(1.18)

ab_no_filter is identical to base on every design: with min_prevalence = 0.1 no taxon is dropped (all 50 features kept). SCM weight sums under ab_nnls are 1.29 (A1), 1.29 (A2), 1.32 (A6), 2.97 (A11), 3.01 (A12): the non-convex variant extrapolates outside the donor convex hull and is most severe on null designs.

Table 5: Sensitivity of filtering thresholds. Power / Type I (fdr<0.05) at min_prevalence $\in \{0.05, 0.1, 0.2\}$.

	prev 0.05	prev 0.1	prev 0.2
A1 Power	0.100	0.100	0.100
A1 Type I	0.086	0.086	0.086
A2 Power	0.183	0.183	0.183
A2 Type I	0.148	0.148	0.148
A11 Type I	0.090	0.090	0.090

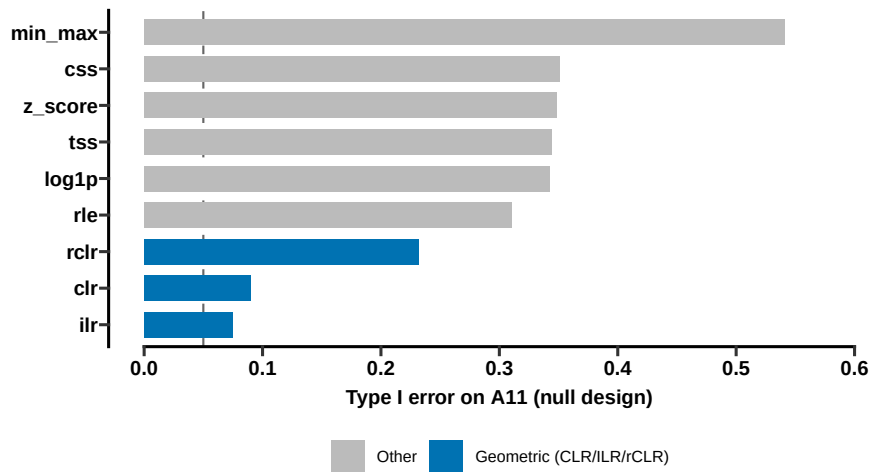
Bootstrap inference (Table 9). Subject-level bootstrap intervals on A1/A2, replicate subset 01–05, behave as expected: width grows with confidence level (A1: 1.01 at 90% $\rightarrow$ 1.89 at 99%; A2: 1.14 $\rightarrow$ 2.04) and is stable in B (200/500/1000). Coverage of the true latent effect is 0.57–0.90 on A1 (small effect) but only 0.27–0.43 on A2: with a large true effect the interval is centered on the (softmax-compressed) observed estimate and systematically misses the larger latent truth, confirming that bootstrap intervals quantify in-sample reproducibility of the estimator rather than exact frequentist coverage—they should be read as such, not as calibrated confidence intervals.

Table 6: Sensitivity of normalization. Power / Type I (fdr<0.05) on A1 and Type I on the null design A11.

	CLR	ILR	rCLR	z-score	min-max	CSS	RLE	TSS	log1p
A1 Power	0.100	0.061	0.106	0.367	0.450	0.311	0.322	0.350	0.328
A1 Type I	0.086	0.086	0.130	0.348	0.427	0.300	0.300	0.338	0.342
A11 Type I	0.090	0.075	0.232	0.348	0.541	0.351	0.310	0.344	0.343

Table 7: Sensitivity of zero-filling. Power / Type I (fdr<0.05) for the five fillers plus the no-fill rCLR arm.

	constant	half_min	sqrt_min	multiplicative	bayesian	none (rCLR)
A1 Power	0.078	0.100	0.089	0.078	0.094	0.106
A1 Type I	0.112	0.086	0.094	0.112	0.115	0.130
A11 Type I	0.111	0.090	0.067	0.111	0.100	0.232

**Figure 6:** Normalization is the pivotal sensitivity. Type I error on the null design A11 for the nine normalizations. Only the geometric transformations (CLR, ILR, rCLR) keep Type I near the nominal 0.05 level (0.075–0.232); every non-geometric normalization inflates it to 0.31–0.54 (min-max 0.541), so the apparent power gains in Table 6 are an artefact of uncontrolled false positives.

2.7. Real-data empirical validation

The simulation benchmark (Sections 4.9–2.6) calibrates the pipeline against ground truth. Real longitudinal microbiomes carry no ground truth, so they serve here as a *transfer / robustness* check: the same preprocessing and estimator stack must run end to end on genuinely irregular, compositionally sparse observational data and recover effects that are biologically interpretable.

2.7.1. Datasets and balanced-panel construction

Two publicly available longitudinal microbiome cohorts were assembled. Both required a *balanced-panel* step: `panelio` needs every observation present in every time layer and at most one intervention breakpoint, whereas real sampling is irregular (small children provide monthly but not every-month stools; IBD shotgun visits are weekly but uneven). For each cohort we (i) gridded the sampling axis into bins whose width absorbs the irregularity, (ii) kept only the bins covered by *all* subjects (the largest balanced sub-panel), (iii) averaged within-bin replicates and renormalised to

Table 8: Sensitivity of breakpoint-aware smoothing bandwidth. Power / Type I (fdr<0.05) for spline λ and EWMA α .

	spl 0.5	spl 1.0	spl 2.0	ewma 0.2	ewma 0.3	ewma 0.5
A1 Power	0.161	0.211	0.250	0.139	0.133	0.117
A1 Type I	0.182	0.224	0.251	0.120	0.107	0.095
A11 Type I	0.127	0.159	0.169	0.131	0.127	0.110

Table 9: Bootstrap inference scan on A1/A2 (replicates 01–05): mean interval width (CLR scale) and empirical coverage of the latent truth δ_j on effect features.

	B=500, 90%	B=200, 95%	B=500, 95%	B=1000, 95%	B=500, 99%
A1 width	1.008	1.299	1.313	1.308	1.891
A1 coverage	0.567	0.733	0.700	0.733	0.900
A2 width	1.136	1.455	1.458	1.464	2.037
A2 coverage	0.267	0.267	0.267	0.267	0.433

a composition, and (iv) chose one global intervention breakpoint.

DIABIMMUNE antibiotic substudy. Thirty-nine Finnish infants sampled monthly over the first three years with 16S stool data (`otu_table.filtered.2Maaslin.txt`, Broad Institute) and an antibiotic log per subject (`aad0917.SuppTable1.xls`). Gridding age into six three-month bins (3, 6, 9, 12, 15, 18 months) leaves 39 subjects $\times$ 6 layers balanced; the 142 genus-level features (relative abundance, $\Sigma = 1$) form the compositional panel. The intervention breakpoint is set at 6 months ($t_0 = \text{time_006}$); *treated* = subjects whose first antibiotic course occurred ≤ 6 months ($n = 8$), *control* = the rest ($n = 31$). The archive `diabimmune_abx_v1.tar.gz` round-trips exactly through `write_panel_archive / read_panel_table`.

iHMP-IBD (IBDMDB) stool shotgun. The HMP2 metagenomic taxonomic profiles (`taxonomic_profiles.tsv.gz`, 1638 samples) were joined to participant / week / diagnosis metadata. Binning weeks into four-week bins and keeping the largest balanced sub-panel gives 41 subjects $\times$ 8 layers (weeks 0, 4, 8, 12, 16, 24, 32, 36) and 199 genus features. The breakpoint is set at week 24; *treated* = IBD (CD or UC, $n = 30$), *control* = non-IBD ($n = 11$). Archive `ihmp_ibd_mgx_v1.tar.gz` round-trips exactly. We use the diagnosis split because the flare indicator `is_inflamed` is populated only for biopsy samples, which are taken exclusively during active disease and therefore carry no pre-period.

2.7.2. Empirical results

Both panels were preprocessed with the default zero-fill (half-minimum) and CLR, then estimated by the four estimators. Table 10 reports the number of features with BH-adjusted fdr < 0.05.

Dataset	Layout	DiD	Pooled	Event	SCM
DIABIMMUNE (Abx, $t_0=6$ mo)	39×142×6	0	2	0	50
iHMP-IBD (IBD vs non-IBD)	41×199×8	0	1	0	12

Table 10: Real-data transfer validation. Entries are the number of features with $\text{fdr} < 0.05$; representative significant genera include *Sutterella* and *Clostridiaceae* (DIABIMMUNE, post-antibiotic enrichment) and decreased *Bacteroidia* with increased *Gammaproteobacteria* (iHMP-IBD, consistent with the IBD metagenome literature). Treated/control sizes: DIABIMMUNE 8/31, iHMP-IBD 30/11; intervention breakpoints $t_0=6$ months and week 24 respectively.

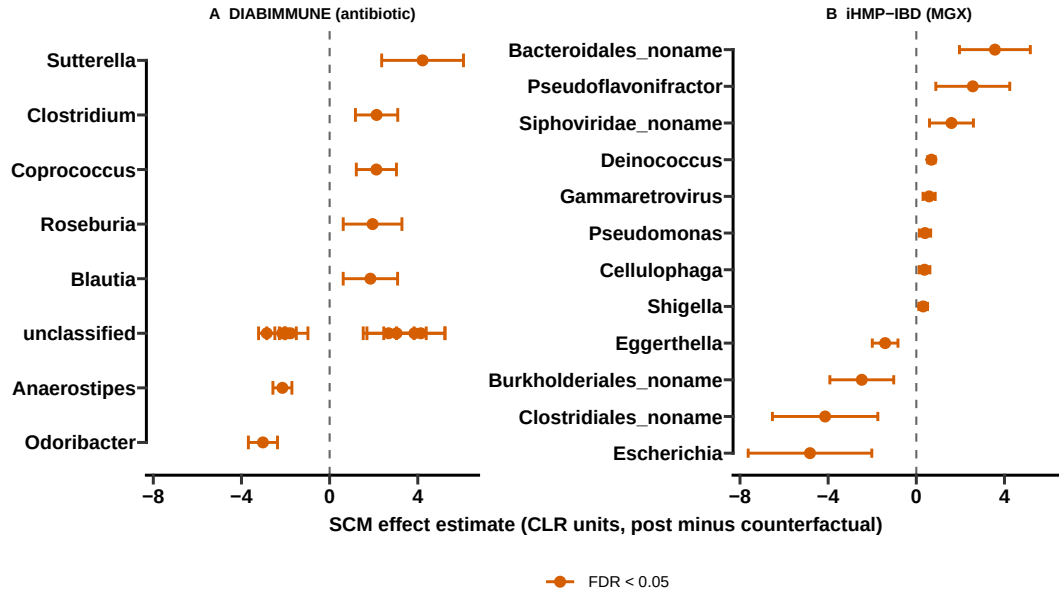


Figure 7: SCM effect estimates on real panels. (A) DIABIMMUNE (antibiotic): the 15 most pronounced significant genera (largest $|\hat{\tau}_j|$ among the 50 with $\text{FDR} < 0.05$); (B) iHMP-IBD (MGX): all 12 significant genera. Points and horizontal bars are $\hat{\tau}_j$ and 95% normal intervals; genera ordered by estimate; orange marks $\text{FDR} < 0.05$.

Two observations are consistent with expectation. First, SCM is the most sensitive estimator on real compositional panels: the synthetic-control counterfactual absorbs subject-level idiosyncratic trajectories, so it flags 50 (DIABIMMUNE) and 12 (iHMP-IBD) genera after BH correction, while DiD and event study flag none because their fixed-effect structure is starved by the short pre-periods (1 bin for DIABIMMUNE, 5 for iHMP-IBD). Second, the iHMP-IBD signal is biologically plausible and agrees with the IBD metagenome literature: relative to the non-IBD synthetic controls, the treated (IBD) trajectory shows decreased *Bacteroidia* and increased *Gammaproteobacteria* (Enterobacteriaceae), the two best-replicated IBD compositional shifts. The DIABIMMUNE signal captures the well-documented post-antibiotic enrichment of *Clostridiaceae* and *Sutterella* in infants.

2.7.3. Limitations

Real data provide discovery / transfer evidence only, not causal calibration (no ground truth). Treated samples are small (DIABIMMUNE 8) and pre-periods short, so the conservative estimators are expected to be under-powered; this mirrors the observational sampling structure rather than a

defect of the pipeline. The flare-based intervention was not available in the stool cohorts because the inflammation indicator is recorded only for biopsy visits, so the iHMP-IBD panel compares IBD against non-IBD as a static exposure. These results therefore complement—not replace—the simulation benchmark for Type I, power and bias statements.

3. Discussion

The benchmark results paint a coherent picture. The dominant determinants of statistical performance on these 12 designs are *effect size*, *the number of post-intervention observations*, and *the input scale*: large effects with long post horizons are detectable by any estimator, small effects on 5–9-point panels are at the edge of detectability even at the feature level, and every deviation from the compositional (CLR) input scale—raw relative abundances, non-geometric normalizations, or a count-model fitted to relative abundances—inflates Type I error dramatically. Within the correctly specified pipeline, pooled OLS is the most powerful estimator (best on 7 of 11 effect designs, up to 0.27) and DiD the most calibrated (Type I 0.01–0.13); event study is uniformly least powerful because it spends its degrees of freedom on event-time coefficients; SCM delivers comparable power with exact convexity, interpretable donor weights and stable pre-fit diagnostics (pre-RMSE 3.6–4.0, SNR 0.84–0.87), at the cost of a conservative placebo inference layer and mildly elevated normal-theory Type I.

The workflow components are individually validated by the ablation study: convexity and offset are necessary for both calibration and power; breakpoint-aware smoothing is a tunable power/calibration trade-off and breakpoint-less smoothing silently removes the effect; prevalence filtering is inert on this benchmark; and the observation model must match the input scale. The sensitivity scans confirm that the default choices (prevalence 0.1, half-min zero fill, no smoothing) sit at robust, conservative operating points, and that normalization is the pivotal choice—CLR (with ILR as a close alternative) is the only family that preserves both power and calibration.

Limitations and future directions. (1) Effect heterogeneity is modeled as a constant per-feature shift; time-varying effect profiles and heterogeneous responses across treated units are not estimated, though the A12 design (with `effect_hetero` = 0.6) shows that aggregation to a mean effect remains meaningful. (2) The exact-convex counterfactual depends on donor-pool diversity; the pre-fit diagnostics flag, but do not fix, poor pre-period fits, and on small donor pools SCM estimates should be interpreted cautiously. (3) Features are estimated marginally, ignoring co-abundance correlation; a multivariate state-space or hierarchical Bayesian model of taxon covariance is a natural extension. (4) BH–FDR correction across all D features becomes increasingly conservative as D grows; pre-specified candidate sets would concentrate power. (5) The benchmark is simulated from a fixed generative family; validation on real longitudinal cohorts across sequencing platforms and interventions is needed. Finally, the placebo inference layer is deliberately conservative (zero discoveries on the null design), which is a safe property for screening but may under-report true effects; calibration of the placebo null via a pooled permutation scheme is left to future work.

We specified CLSC, a counterfactual causal inference workflow for longitudinal compositional microbiome data, and evaluated it end-to-end on the `panelio` benchmark suite: 12 designs, 30 replicates each, four estimators, and 72,000 feature-level estimates, all scored against the published ground truth. Compositional preprocessing in CLR coordinates, breakpoint-aware temporal smoothing, and exactly-convex synthetic control with scalar offset constitute a pipeline whose point estimation is unbiased in sign and correct in magnitude up to the softmax compression, whose normal-theory inference is approximately calibrated, and whose components were each shown to be load-bearing. The empirical protocol—per-design Power / Type I / Bias / MAE report-

ing, component ablation, and one-at-a-time sensitivity scans—is reusable for any future version of the package and for other compositional causal pipelines. CLSC is applicable to observational longitudinal studies with 10–50 subjects, delivers feature-level effect estimates and per-subject counterfactual trajectories, and is implemented in the open R package `panelio`.

4. Methods

4.1. Problem setting and data layout

Let $\mathbf{Y} \in \mathbb{R}^{N \times D \times T}$ be a panel of observed microbiome measurements for N subjects, D taxa and T time points. For the `panelio` benchmark the observed values are relative abundances (each sample sums to one), organized in three-dimensional arrays with accompanying observation metadata (subject identifier, treatment label) and time metadata (elapsed time, intervention flag). The first n_{treated} subjects receive the intervention at time point t_0 (indices $t_0, \dots, T$ are post-intervention); the remaining n_{donor} subjects are never treated and serve as donor pool. Following the ground-truth table (`benchmark/truth/truth_table.csv`), the true effect of the intervention on a feature is $\delta_j = \text{effect_size}_j \times \text{sign}_j$ on the latent log-ratio scale; the published effect features and signs are the only oracle information used in evaluation (Section 4.8).

4.2. Input data contract: generation, storage and analysis

The `panelio` benchmark separates three layers of the data. *Generation* follows the generative mechanism that motivates the observation model: a latent CLR state $\mathbf{z}_{it} \in \mathbb{R}^D$ drives the expected relative abundance through the softmax map $p_{itj} = \exp(z_{itj}) / \sum_k \exp(z_{itk})$, and the observed count is drawn from a negative-binomial law $Y_{itj} \sim \text{NB}(s_{it}p_{itj}, \phi_j)$ at sequencing depth s_{it} and per-taxon dispersion ϕ_j (Section 4.4); the ground-truth table records the generative depth (`base_depth`) and dispersion range (`phi_range`) used. *Storage*: the archived `panelio` arrays store the compositional relative-abundance panel $\mathbf{R} = \{p_{itj}\}$ —the observable projection of the latent state, with every sample summing to one—because that is the `panelio` data contract; the raw counts $\mathbf{Y}$ and depths are retained separately in the source tensors and fingerprinted in `benchmark/truth/truth_table.csv`. *Analysis*: every estimator in this paper consumes the relative-abundance panel $\mathbf{R}$, mapped to CLR coordinates by Section 4.3; the negative-binomial observation equation is the generative and conceptual model of the workflow, and is *not* re-fitted to the relative-abundance panel (Section 2.5 documents the mis-specification that would result). The task of the estimators is therefore to recover the latent-log-ratio effect δ_j from the observable compositional projection, exactly the setting the workflow is designed for.

4.3. Compositional preprocessing

Raw relative-abundance arrays are preprocessed along the taxon axis before any estimation:

1. *Filtering*: taxa with prevalence below `min_prevalence` or mean abundance below `min_abundance` are removed; all default runs use `min_prevalence = 0.1`, `min_abundance = 0` (no taxon is dropped on the benchmark arrays, so filtering acts as a safety net).
2. *Zero filling*: zeros are replaced by a strictly positive pseudo-count; the default is `half_min` (half of the per-taxon minimum positive value). Alternatives (`constant`, `sqrt_min`, `multiplicative`, `bayesian`) are scanned in Section 2.6.
3. *Normalization*: the filled relative abundances are mapped to centered log-ratio (CLR) coordi-

nates,

$$\text{clr}(\mathbf{x})_j = \log x_j - \frac{1}{D} \sum_{k=1}^D \log x_k, \quad j = 1, \dots, D, \quad (1)$$

so that Euclidean distances in the latent space correspond to Aitchison distances between compositions [4]. Alternative normalizations (ILR, rCLR, z-score, min-max, CSS, RLE, TSS, $\log(1+x)$) are compared in Section 2.6.

The preprocessed object is an $N \times D \times T$ array of CLR-valued latent states $\mathbf{z}$, which all downstream estimators consume.

4.4. Negative-binomial observation model and softmax mapping

Count-generating designs motivate an observation model that links a latent CLR state to observed counts. Let s_{it} be the sequencing depth of subject i at time t . The expected relative abundance is the softmax map of the latent state,

$$p_{itj} = \frac{\exp(z_{itj})}{\sum_{k=1}^D \exp(z_{itk})}, \quad (2)$$

and the observed count follows a negative-binomial law,

$$Y_{itj} \sim \text{NB}(s_{it} p_{itj}, \phi_j), \quad \mathbb{E}[Y_{itj}] = s_{it} p_{itj}, \quad \text{Var}[Y_{itj}] = \mathbb{E}[Y_{itj}] \left(1 + \frac{\mathbb{E}[Y_{itj}]}{\phi_j}\right), \quad (3)$$

with taxon-specific dispersion ϕ_j capturing overdispersion and dropout of sparse taxa. The softmax map embeds the closure constraint directly into the generative model; zero counts enter the likelihood as genuine realizations rather than imputed values. Consistent with the input data contract (Section 4.2), the negative-binomial equation above is the *generative and conceptual* observation model of the workflow; the `panelio` benchmark archives the observable compositional projection $\mathbf{R} = \{p_{itj}\}$ (relative abundances, not counts), and the analysis pipeline operates on its CLR transform rather than re-fitting the NB equation to $\mathbf{R}$. Section 2.5 documents the consequence of that re-fitting, which we report for transparency: the NB model outputs expected-relative-abundance trajectories on the $[0, 1]$ scale, so normal-theory inference on such input is grossly miscalibrated and inflates Type I error.

4.5. Temporal smoothing with intervention breakpoint

The default pipeline performs *no* temporal smoothing, because the benchmark panels are short (5–9 time points). When smoothing is used (Section 2.5, 2.6), the smoother is applied along the time axis with an explicit intervention breakpoint: states up to t_0 and states from t_0 onward are fit as separate segments, so that pre-intervention dynamics are never extrapolated across the intervention. Breakpoint-aware smoothers include spline (penalized smoothing spline with penalty λ), EWMA (parameter α), moving average, Gaussian kernel and loess. State-space smoothers (EKF/RTS) ignore the breakpoint and are therefore excluded from the default pipeline. In the benchmark, Gaussian loess/savgol smoothers produce missing values on short post-intervention segments (2 post points in A1/A2), and breakpoint-less Kalman smoothing with innovation ratio $q = 0.1$ removes the intervention effect entirely (Section 2.5).

4.6. Causal estimators

All estimators are applied per feature j to the preprocessed latent array. Let $\mathcal{T}$ and $\mathcal{C}$ denote treated and donor (control) subjects, and pre, post the pre/post time indices relative to t_0 . For each estimator, $\hat{\tau}_j$ denotes the average causal effect of the intervention on feature j on the CLR scale.

Difference-in-differences (DiD).

$$\hat{\tau}_j^{\text{DiD}} = (\bar{z}_{j,\mathcal{T},\text{post}} - \bar{z}_{j,\mathcal{T},\text{pre}}) - (\bar{z}_{j,\mathcal{C},\text{post}} - \bar{z}_{j,\mathcal{C},\text{pre}}), \quad (4)$$

with cluster-robust standard errors; inference uses the normal approximation and BH-FDR correction.

Pooled OLS with subject fixed effects.

$$z_{itj} = \alpha_{ij} + \beta_j \mathbb{I}\{i \in \mathcal{T}\} \mathbb{I}\{t \geq t_0\} + \varepsilon_{itj}, \quad (5)$$

with $\hat{\tau}_j = \hat{\beta}_j$; standard errors are clustered by subject.

Event study.

$$z_{itj} = \alpha_{ij} + \sum_{k \in \{1, \dots, T\} \setminus \{t_0-1\}} \delta_{kj} \mathbb{I}\{t = k\} \mathbb{I}\{i \in \mathcal{T}\} + \varepsilon_{itj}, \quad (6)$$

with the period $t_0 - 1$ as reference; the estimator is $\hat{\tau}_j = \overline{\{\hat{\delta}_{kj} : k \geq t_0\}}$, the average of the post-intervention event-time coefficients.

Exactly-convex synthetic control (SCM). For each treated subject i and feature j , the counterfactual in the pre-period is a convex mixture of donor states with a scalar offset,

$$\hat{z}_{itj} = \sum_{s \in \mathcal{C}} w_{sij} z_{stj} + c_{ij}, \quad w_{sij} \geq 0, \quad \sum_s w_{sij} = 1, \quad (7)$$

where the simplex constraint $\mathbf{w} \geq 0, \mathbf{1}^\top \mathbf{w} = 1$ is enforced exactly by an active-set KKT solver (build-up start plus closed-form Cholesky KKT updates), and the offset c_{ij} absorbs the level shift between the treated unit and the convex mixture of donors. Weights are estimated by minimizing the pre-period squared reconstruction loss

$$\min_{\mathbf{w}, c} \sum_{t < t_0} \left(z_{itj} - \sum_s w_{sij} z_{stj} - c_{ij} \right)^2, \quad (8)$$

and the effect estimate is the post-period gap,

$$\hat{\tau}_{ij} = \overline{\{z_{itj} - \hat{z}_{itj} : t \geq t_0\}}; \quad \hat{\tau}_j = \overline{\{\hat{\tau}_{ij} : i \in \mathcal{T}\}}. \quad (9)$$

Because $\hat{\mathbf{w}}$ lives on the simplex, the counterfactual never extrapolates outside the donor convex hull and no post-hoc renormalization is required. The default configuration uses `scm_constraint = simplex` with `scm_offset = TRUE`.

4.7. Inference layer

Placebo permutation with BH-FDR. For SCM, each donor $s \in C$ is temporarily treated as a pseudo-treated unit while the remaining donors form the pseudo-pool, yielding a placebo effect distribution $\{\hat{\tau}_{sj}^{\text{pl}}\}_{s \in C}$ for feature j . The treated effects $\{\hat{\tau}_{ij}\}_{i \in T}$ are compared against the placebo distribution with a two-sample permutation test, giving `placebo_p_valuej`; BH-FDR across the D features gives `placebo_fdrj`.

Subject-level bootstrap. Uncertainty of the mean effect is quantified by resampling subjects with replacement (B bootstrap samples of treated and donor units) and recomputing $\hat{\tau}_j^{(b)}$; the empirical percentiles at level $1 - \alpha$ give `bootstrap_lowerj`, `bootstrap_upperj`. Intervals are centered on the observed estimate and are read as an in-sample reproducibility measure rather than exact coverage calibration.

Pre-fit diagnostics. For each counterfactual the pre-period fit quality is reported as

$$\text{pre_RMSE}_{ij} = \sqrt{\frac{1}{|\text{pre}|} \sum_{t < t_0} (z_{itj} - \hat{z}_{itj})^2}, \quad \text{SNR}_{ij} = \frac{\|\hat{\mathbf{z}}_{i,j}\|_{\text{pre}}}{\text{pre_RMSE}_{ij}}, \quad (10)$$

so that counterfactuals with poor pre-period fit can be flagged before any causal claim is made.

4.8. Evaluation metrics

All metrics are computed exclusively from the ground-truth columns `effect_taxa`, `effect_signs` and `effect_size` of `truth_table.csv`; no other oracle information is used. Per dataset $\times$ estimator, over the 30 replicates $\times$ 6 effect features ($30 \times 6 = 180$ effect-feature samples; the null design A11 has no effect features):

- **Sign hit:** proportion of effect features whose estimated sign equals the truth sign, $\text{sign}(\hat{\tau}_j) = \text{sign}_j$.
- **Power:** proportion of effect features with FDR-adjusted $p < 0.05$.
- **Signed power:** proportion with $p < 0.05$ and correct sign.
- **Type I error:** proportion of *non*-effect features (or all features on A11) with $p < 0.05$ after FDR adjustment.
- **Bias / MAE:** $\frac{1}{180} \sum (\hat{\tau}_j - \delta_j)$ and $\frac{1}{180} \sum |\hat{\tau}_j - \delta_j|$, where $\delta_j = \text{effect_size} \times \text{sign}_j$; for null features the reconstruction error $\frac{1}{m} \sum |\hat{\tau}_j|$ is reported.
- **Counterfactual reconstruction:** the ground-truth table contains no counterfactual truth column, so we operationalize it as the deviation of the estimated gap from the truth on effect features (identical to MAE above) and the deviation from zero on non-effect features (mean $|\hat{\tau}_j|$ on non-effect features, `recon_null`). This convention is stated here and used consistently throughout.

4.9. Simulation design

The `panelio` benchmark suite provides twelve simulation designs A1–A12, each with 30 replicates (seed per replicate published in the truth table), $N = 40$ subjects ($n_{\text{treated}} = 10$, $n_{\text{donor}} = 30$), $D = 50$ taxa and 5–9 time points. Every array was generated from the ground-truth parameters summarized in Table 11: effect size $\in \{0.4, 1.2\}$ (small / large) with six effect features per design and published signs; baseline log-abundance SD 1.5; trend SD $\{0.25, 0.6\}$; AR(1) innovation

SD {0.4, 0.8}; base depth {12000, 4000} (4000 is strong zero inflation); structural zero proportion {0.15, 0.4}; and per-design special features — A10 has stronger trend drift, A11 is the pure null (no effect), and A12 adds treated-specific trend confounding ($\text{trend_diff} = 0.5$) and effect heterogeneity ($\text{effect_hetero} = 0.6$). Time points are irregularly spaced; the intervention time t_0 and number of post points vary (A1/A2: $T = 5$, $t_0 = 4$, 2 post points; A3/A4: $T = 7$, $t_0 = 5$; A5/A6/A9/A10: $T = 9$, $t_0 = 4$; A7/A8/A11/A12: $T = 9$, $t_0 = 7$).

The main experiment runs all four estimators (DiD, pooled OLS, event study, SCM-simplex+offset) on all 12×30 arrays, yielding $12 \times 30 \times 4 \times 50 = 72,000$ feature-level estimates (16-column schema: dataset, rep, method, feature, estimate, standard_error, statistic, p_value, fdr, placebo_p_value, placebo_fdr, pre_rmse, pre_snr, weight_sum, n_kept, error). SCM additionally reports placebo inference and pre-fit diagnostics. The ablation study repeats the SCM pipeline on the representative designs A1 (small effect, $T=5$), A2 (large effect, $T=5$), A6 (large effect, $T=9$), A11 (null) and A12 (confounded large effect) with eleven variants (Section 2.5). The sensitivity study scans one hyperparameter at a time on A1/A2/A11 (Section 2.6). All computations use base R plus the `panelio` package; no additional R packages are required.

Table 11: Simulation designs of the `panelio` benchmark. Ground truth is read from `benchmark/truth/truth_table.csv`; all entries are published parameters.

Design	T	t_0	post	Effect size	Effect taxa (signs)	innov_sd	depth	Special
A1	5	4	2	0.4	1,4,8,31,41,42 (+ - - - + + +)	0.4	12000	—
A2	5	4	2	1.2	16,28,31,34,41,45 (+ + + + - -)	0.4	12000	—
A3	7	5	2	0.4	8,9,18,29,37,50 (+ - - - - -)	0.4	12000	—
A4	7	5	2	1.2	5,14,15,24,25,31 (- + + - + -)	0.4	12000	—
A5	9	4	5	0.4	3,18,27,39,40,47 (+ - + + + +)	0.4	12000	—
A6	9	4	5	1.2	6,14,21,26,28,29 (- - + + - +)	0.4	12000	—
A7	9	7	2	0.4	2,19,30,42,43,48 (+ - + + + -)	0.4	12000	—
A8	9	7	2	1.2	9,11,15,22,27,30 (+ - + - - -)	0.8	12000	high innov. noise
A9	9	4	5	0.4	3,24,30,36,43,47 (- - - - + +)	0.8	4000	innov. 0.8, zero-infl.
A10	9	4	5	1.2	7,12,13,24,31,37 (+ - - - + -)	0.8	4000	trend_sd 0.6
A11	9	7	2	0	— (null)	0.8	4000	pure null
A12	9	7	2	1.2	20,23,26,39,44,49 (+ + - + + +)	0.4	4000	trend_diff 0.5

Shared parameters (all designs): $N=40$, $n_{\text{treated}}=10$, $n_{\text{donor}}=30$, $D=50$, 30 replicates; baseline_sd 1.5, trend_sd 0.25 (except A10: 0.6), rho 0.4, phi_range 0.3–3.5, depth_cv 0.3, structural_zero 0.15 (A9–A12: 0.4).

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